

Applied Optimization

Exercise 11

Interior Point Method & Algorithms Overview

What to hand in

A .zip compressed file renamed to `Exercisen-Group.zip` where n is the number of the current exercise sheet. It should contain:

- Hand in **only** the files you changed (headers and source). It is up to you to make sure that all files that you have changed are in the zip.
- A `readme.txt` file containing a description on how you solved each exercise (use the same numbers and titles) and the encountered problems. Indicate what fraction of the total workload each project member contributed.
Failure to do so will result in you losing a point.
- Other files that are required by your `readme.txt` file. For example, if you mention some screenshot images in `readme.txt`, these images need to be submitted too.
- For the theory exercise, put `TheoryExercise.pdf` with your solutions in the same .zip you submit for the code.
- Submit your solutions to ILIAS before the submission deadline. Late submissions will receive 0 points!
- **Important:** Your submission MUST compile without any issue in order to obtain a passing grade.

Algorithms Overview (1 pt)

Consider the task of optimizing a **large-scale** problem with $n = 10^6$ variables, given that state-of-the-art implementations of the following methods are available:

- (a) Gradient Descent method
- (b) Newton's method as on Page 12, Slides-07
- (c) Augmented Lagrangian method using Newton's solver
- (d) Infeasible Start Newton's method
- (e) Active Set Method using Newton's solver
- (f) Primal-Dual Interior Point method using Newton's solver with Hessian modification
- (g) Gauss-Newton method
- (h) L-BFGS method

For every of the following scenarios, list algorithms that can find a local minimum and specify the algorithm for which you expect the shortest runtime (assuming infinite memory). Note that every problem starts from a feasible point. **Briefly justify your choice.**

1. A problem which is strictly convex with linear equality constraints.
2. A linear program with inequality constraints.
3. A convex quadratic problem with linear inequality constraints which can be solved via a sequence of convex quadratic subproblems with small number of linear equality constraints.
4. An unconstrained non-convex problem
5. A non-convex least-squares problem of the form $f(x) = \sum_{i=1}^m f_i(x)^2$ with $x \in \mathbb{R}^n$. The gradient of f_i is dense and the hessian is not available.

Solution:

1. It is an equality constrained strictly convex problem. So c, d, e, f are able to solve it. Best: d, e, f. Augmented Lagrangian method needs iterations of the outer loop to adjust the penalty parameter and dual variables, while the other three methods don't. d, e, f are identical in this case.

2. A linear program with inequality constraints has zero hessian matrix. The only choice is f which modifies the hessian to be positive definite.

3. If the sequence is already given, then we can use any method to solve each sub-problem individually. If it's only *possible* to obtain one, but not provided, then the Active Set Method is best suited to solve this problem. Indeed, it will basically generate this generate this sequence.

4. Since the problem is non-convex, a, f, h work for non-convex problems. g would also work **if** the objective function can be expressed as a sum of squares. f is equipped with Newton's solver with hessian modification that can handle indefiniteness of the hessian matrix. So it would perform the best.

5. The hessian is not available, so a, g, h which don't use second order information would work. Since the gradient is dense, Gauss-Newton is not a good option because the approximated hessian will be dense too. So here we choose h as the best.